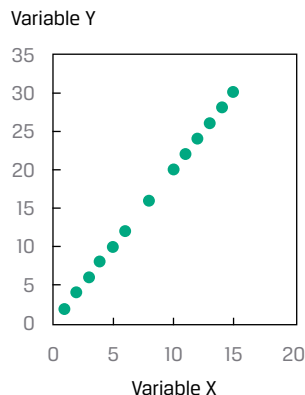


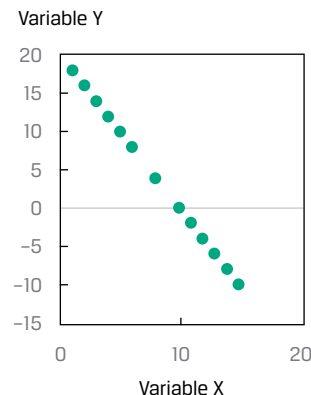
(but positive), the correlation between the two variables would still be $+1$ as long as all the points lie on that straight line. Panel B shows a scatter plot for two variables with a correlation coefficient of -1 . Once again, the plotted observations all fall on a straight line. In this graph, however, the line has a negative slope. As X increases by one unit, Y decreases by two units, regardless of the initial value of X .

Exhibit 26: Scatter Plots Showing Various Degrees of Correlation

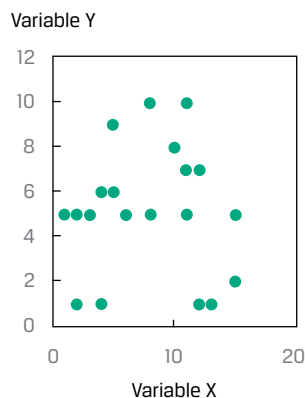
A. Variables With a Correlation of $+1$



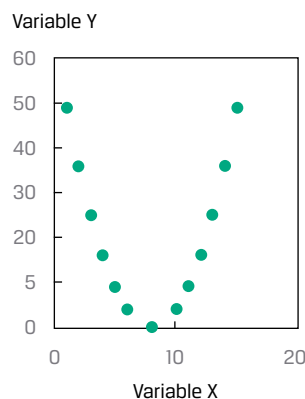
B. Variables With a Correlation of -1



C. Variables With a Correlation of 0



D. Variables With a Strong Nonlinear Association



Panel C shows a scatter plot of two variables with a correlation of 0 ; they have no linear relation. This graph shows that the value of variable X tells us nothing about the value of variable Y . Panel D shows a scatter plot of two variables that have a non-linear relationship. Because the correlation coefficient is a measure of the linear association between two variables, it would not be appropriate to use the correlation coefficient in this case.

Limitations of Correlation Analysis

Exhibit 26 illustrates that correlation measures the linear association between two variables, but it may not always be reliable. Two variables can have a strong *nonlinear* relation and still have a very low correlation. A nonlinear relation between variables X and Y is shown in Panel D. Even though these two variables are perfectly associated, there is no linear association between them and hence no meaningful correlation.